

# Product Specification Document

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## SmartHydro – Intelligent Water Management System

### 1. Product Overview

SmartHydro is a connected water management system designed to help households and small businesses monitor, control, and optimize their water consumption. The system combines smart sensors, a central control unit, and a mobile application to provide real-time insights into water usage and detect leaks early.

### 2. Objectives

- Reduce water waste
- Detect leaks and anomalies early
- Provide real-time consumption insights
- Enable remote control of water flow

### 3. System Components

- Smart Flow Sensors: Installed on water pipes to measure flow and detect anomalies.
- Central Hub: Collects data from sensors and communicates with the cloud.
- Mobile Application: User interface for monitoring and control.
- Cloud Backend: Stores data and provides analytics services.

### 4. Key Features

- Real-time water usage tracking
- Leak detection alerts
- Historical data and analytics
- Remote water shut-off control
- Multi-user access with role-based permissions

### 5. Technical Architecture

The system uses IoT sensors connected via Wi-Fi or LoRaWAN to a central hub. The hub communicates with a cloud platform through secure HTTPS connections. The mobile app interacts with the backend via REST APIs.

### 6. Security & Compliance

- End-to-end encryption (TLS)
- Secure authentication (OAuth 2.0)
- GDPR-compliant data handling
- Role-Based Access Control (RBAC)

## 7. Use Cases

- Homeowners monitoring daily water usage
- Property managers detecting leaks across multiple units
- Small businesses optimizing operational costs